

IDEA SUBMISSION — TITLE PAGE

Problem Statement — Build a working voice agent where Rime-generated speech drives the main interaction and Qdrant gives the system a real job (knowledge, memory, or routing) — not a chatbot with a microphone bolted on.

Round — Round 1 (Online Submission)

Track: VoxForge

Team Name (Registered on portal) — VectorVoice

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ATHENAEUM — “YOUR LIBRARY, NOW YOU CAN TALK TO IT.”

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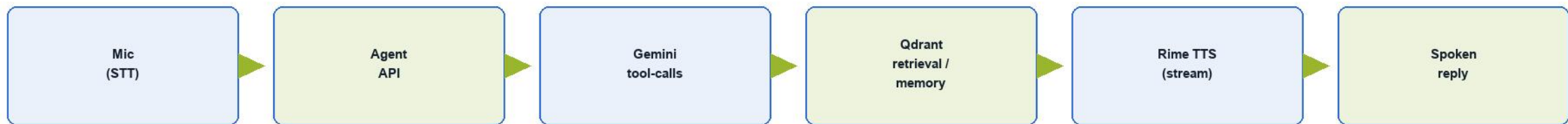
Proposed Solution (Describe your Idea / Solution / Prototype)

- **Detailed explanation:** A voice-first AI library assistant — patrons speak naturally (“find me something beginner-friendly about AI”) instead of typing exact titles/keywords. Rime speaks every response back; the whole loop is voice-in, voice-out.
- **How it addresses the problem:** Rime (Mist/Coda) drives 100% of spoken output — not a decorative add-on. Qdrant does three real jobs: semantic book retrieval, similar-book recommendation, and per-user preference memory, each isolated by a userId payload filter.
- **Innovation and uniqueness:** Barge-in interruption (user can talk over the assistant mid-sentence and it stops instantly), a grounded tool-calling agent (Gemini) that never invents a book, and a memory system so “remember I like sci-fi” actually changes future recommendations.

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- **Technologies:** Next.js 15 (TypeScript, App Router) full-stack; Rime REST streaming TTS (Mist v3 model, server-side only — key never touches the client); Qdrant Cloud (768-dim, Cosine) for two collections — library_books and user_memory; Gemini 2.5 (tool-calling agent + gemini-embedding-001); browser-native Web Speech API for STT.

Methodology / process for implementation:



- **Barge-in:** Web Speech API's onSpeechStart fires the instant voice energy is detected; if the assistant is speaking, playback + fetch are aborted immediately and state flips to listening.
- **Grounded agent loop:** Gemini picks from 6 tools (search_books, get_book, recommend_books, check_availability, get_user_memory, save_user_memory) and replays function-call turns verbatim (including thought_signature) so multi-round tool chains stay valid.
- **Demo/live duality:** Every external call (Rime / Qdrant / Gemini) is wrapped so a missing key or failed call falls back to deterministic demo data, tagged mode: "live" | "demo" — never silently fakes a working integration.

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- **Feasibility — already working end-to-end:** Live production deployment on Vercel with real Rime audio streaming, real Qdrant semantic search over 142 embedded books, and a real Gemini tool-calling agent — this is not a mockup.
- **Potential challenges:** (1) Gemini free-tier daily quota (20 req/day) can throttle heavy demo use; (2) Qdrant requires explicit payload indexes per filtered field or filtered queries error out; (3) newer Gemini models require replaying the exact thought_signature on multi-turn tool calls or the next call is rejected; (4) Web Speech API only works reliably in Chromium browsers.
- **Strategies for overcoming these:** Automatic, visible demo-mode fallback (keyword search + rule-based routing) keeps the product always-demoable even mid-quota-exhaustion; ensureMemoryUserIdIndex() self-heals the Qdrant index on first use; the orchestrator replays the model's raw response content verbatim instead of reconstructing it; the UI detects unsupported browsers and shows a clear message instead of failing silently.

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- **Potential impact on the target audience:** Library patrons who don't know exact titles/authors (students, casual readers, visually impaired users) can find and borrow books through natural conversation instead of keyword search — voice removes the “you have to already know what you're looking for” barrier of a traditional catalog.
- **Accessibility:** Hands-free, screen-free interaction benefits visually impaired patrons and anyone for whom typing is a barrier; the assistant reads locations, availability, and recommendations aloud with grounded, explainable reasoning (“I recommended this because...”).
- **Economic / operational:** Reduces librarian time spent on repetitive “do you have a book about...” queries; scales to any library's real catalog by re-running the ingestion pipeline against their own metadata.
- **Trust and continuity:** Personalized memory (“remember I like sci-fi under 300 pages”) persists across sessions, so recommendations improve over time instead of starting from zero on every visit — directly addressing StarForge's “never start from zero” mission.